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clc
clear
close all

filename = 'Data.csv';
delimiter = ',';
startRow = 3;
formatSpec = '%f%f%f%f%f%f%f%f%f%f%[\n\r]';

fileID = fopen(filename, 'r');

dataArray = textscan(fileID, formatSpec, 'Delimiter', delimiter, 'HeaderLines',
startRow-1, 'ReturnOnError', false);

fclose(fileID);

Sample = dataArray(:, 1);
x1 = dataArray(:, 2);
x2 = dataArray(:, 3);
x3 = dataArray(:, 4);
x4 = dataArray(:, 5);
x5 = dataArray(:, 6);
x6 = dataArray(:, 7);
x7 = dataArray(:, 8);
x8 = dataArray(:, 9);

clearvars filename delimiter startRow formatSpec fileID ans;

Data = [ones(10, 1), x1, x2; ones(10, 1), x3, x4; ones(10, 1), x5, x6; ones(10,
1), x7, x8]';

tW = [];
for s = 1:4
    hold on;
    switch s
        case 1
            x = [Data(:, 1:10), Data(:, 11:end)];
            str = 'Class 1 vs Other Classes';
        case 2
            x = [Data(:, 11:20), Data(:, [1:10, 21:end])];
            str = 'Class 2 vs Other Classes';
        case 3
            x = [Data(:, 21:30), Data(:, [1:20, 31:end])];
            str = 'Class 3 vs Other Classes';
        case 4
            x = [Data(:, 31:end), Data(:, 1:30)];
            str = 'Class 4 vs Other Classes';
    end

    fig1 = figure(1);
    axis square
    xlim([-12, 12]);
    ylim([-12, 12]);
    plot(x(2, 1:10), x(3, 1:10), 'or', 'MarkerFaceColor', 'r');
    plot(x(2, 11:end), x(3, 11:end), 'ob', 'MarkerFaceColor', 'b');
    title(str);

    rnd = @(a, b)(a + (b - a) .* rand());

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a = -0.01;
b = +0.01;
K = 2;
d = 2;
N = 40;
c = 0.05;
eta = c;
w = zeros(K, d+1);
delta_w = zeros(K, d+1);
o = zeros(K, 1);
y = zeros(K, 1);
r = zeros(K, N);
Y = zeros(K, N);
r(1, 1:10) = 1;
r(2, 11:end) = 1;

for i = 1:K
    for j = 1:d + 1
        w(i, j) = rnd(a, b);
    end
end

iter = 1;
maxIter = 10000;
while true
    wOld = w;
    for i = 1:K
        for j = 1:d + 1
            delta_w(i, j) = 0;
        end
    end
    for t = 1:N
        for i = 1:K
            o(i) = 0;
            for j = 1:d + 1
                o(i) = o(i) + w(i, j) .* x(j, t);
            end
        end
        o = o - max(o);
        for i = 1:K
            y(i) = exp(o(i)) / sum(exp(o(:)));
        end
        for i = 1:K
            for j = 1:d + 1
                delta_w(i, j) = delta_w(i, j) + (r(i, t) - y(i)) .* x(j, t);
            end
        end
        Y(:, t) = y(:);
    end
    for i = 1:K
        for j = 1:d + 1
            w(i, j) = w(i, j) + eta .* delta_w(i, j);
        end
    end

    wNew = w;
    wb = w(1, :) - w(2, :);

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        if norm(wNew(:) - wOld(:), 2) / max(1, norm(wOld(:), 2)) < 1e-6 || iter >=
maxIter || any(isnan(wNew(:))) || any(isinf(wNew(:)))
            break;
        end
        disp(w);
        iter = iter + 1;
        eta = c / iter;
    end

    plt0 = ezplot([num2str(wb(2)), '* x1 +', num2str(wb(3)), '* x2 +',
num2str(wb(1))], [-20, 20]);
    set(plt0, 'Color', 'magenta', 'LineWidth', 1);
    saveas(fig1, ['plot_2_', num2str(s), '.png']);

    fig2 = figure(2);
    e = bar3(Y);
    for k = 1:length(e)
        zdata = e(k).ZData;
        e(k).CData = zdata;
        e(k).FaceColor = 'interp';
    end
    title(str);
    saveas(fig2, ['bar3_2_', num2str(s), '.png']);

    pause(0.25);
    tW = [tW; wb];

    close all;
end

fig3 = figure(3);
hold on;
axis square
xlim([-12, 12]);
ylim([-12, 12]);
plot(Data(2, 1:10), Data(3, 1:10), 'or', 'MarkerFaceColor', 'r');
plot(Data(2, 11:20), Data(3, 11:20), 'og', 'MarkerFaceColor', 'g');
plot(Data(2, 21:30), Data(3, 21:30), 'ob', 'MarkerFaceColor', 'b');
plot(Data(2, 31:40), Data(3, 31:40), 'oc', 'MarkerFaceColor', 'c');
plt1 = ezplot([num2str(tW(1, 2)), '* x1 +', num2str(tW(1, 3)), '* x2 +',
num2str(tW(1, 1))], [-12, 12]);
set(plt1, 'Color', 'magenta', 'LineWidth', 1);
plt2 = ezplot([num2str(tW(2, 2)), '* x1 +', num2str(tW(2, 3)), '* x2 +',
num2str(tW(2, 1))], [-12, 12]);
set(plt2, 'Color', 'magenta', 'LineWidth', 1);
plt3 = ezplot([num2str(tW(3, 2)), '* x1 +', num2str(tW(3, 3)), '* x2 +',
num2str(tW(3, 1))], [-12, 12]);
set(plt3, 'Color', 'magenta', 'LineWidth', 1);
plt4 = ezplot([num2str(tW(4, 2)), '* x1 +', num2str(tW(4, 3)), '* x2 +',
num2str(tW(4, 1))], [-12, 12]);
set(plt4, 'Color', 'magenta', 'LineWidth', 1);
title('');
saveas(fig3, 'plot_3.png');

g = linspace(-12, 12, 300);
[XX, YY] = meshgrid(g, g);
LABEL = zeros(size(XX));

for i = 1:length(g)

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for j = 1:length(g)
    pos = 0;
    cls = 0;

    for s = 1:4
        val = tw(s, 1) + tw(s, 2) * XX(i, j) + tw(s, 3) * YY(i, j);

        if val > 0
            pos = pos + 1;
            cls = s;
        end
    end

    if pos == 1
        LABEL(i, j) = cls;
    elseif pos == 0
        LABEL(i, j) = 5;
    else
        LABEL(i, j) = 6;
    end
end
end

fig4 = figure(4);
hold on;
axis square
xlim([-12, 12]);
ylim([-12, 12]);

imagesc(g, g, LABEL);
set(gca, 'YDir', 'normal');

colormap([
    1.0 0.75 0.75
    0.75 1.0 0.75
    0.75 0.75 1.0
    0.75 1.0 1.0
    0.75 0.75 0.75
    0.35 0.35 0.35
]);

caxis([1 6]);

for s = 1:4
    plt = ezplot([num2str(tw(s, 2)), '* x1 +', num2str(tw(s, 3)), '* x2 +',
num2str(tw(s, 1))], [-12, 12]);
    set(plt, 'Color', 'magenta', 'LineWidth', 1);
end

plot(Data(2, 1:10), Data(3, 1:10), 'or', 'MarkerFaceColor', 'r');
plot(Data(2, 11:20), Data(3, 11:20), 'og', 'MarkerFaceColor', 'g');
plot(Data(2, 21:30), Data(3, 21:30), 'ob', 'MarkerFaceColor', 'b');
plot(Data(2, 31:40), Data(3, 31:40), 'oc', 'MarkerFaceColor', 'c');

title('Ambiguous Regions for Class vs Other Classes');
saveas(fig4, 'ambiguous_1_vs_others.png');

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